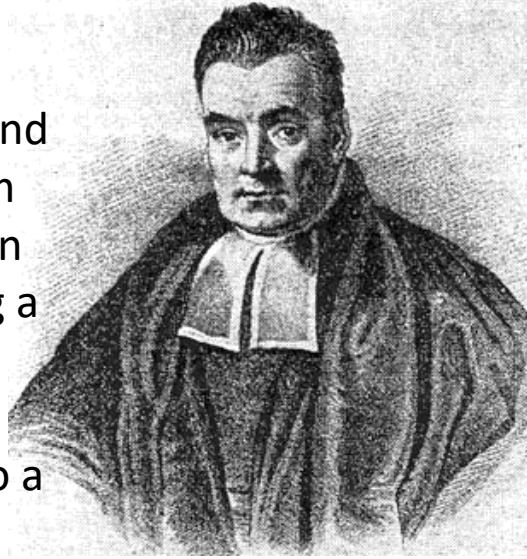
BAYES THEOREM

Bayes' theorem

Bayes' theorem is named after Thomas Bayes, a minister, statistician, and philosopher. Bayes used conditional probability to provide an algorithm (his proposition 9) that uses evidence to calculate limits on an unknown parameter. His work was published in 1763 as an essay towards solving a problem in the Doctrine of Chances. Bayes studied how to compute a distribution for the probability parameter of a binomial distribution (in modern terminology). After Bayes's death, his family gave his papers to a friend, the minister, philosopher, and mathematician Richard Price.



Bayes' Theorem describes the probability of an event based on prior knowledge of conditions related to the event. It is expressed mathematically as:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$ is the probability of event A occurring given that B is true.
- $P(B|A)$ is the probability of event B occurring given that A is true.
- $P(A)$ is the probability of event A occurring.
- $P(B)$ is the probability of event B occurring.

Concept learning

Concept learning is the process of inferring a function fro
training data in supervised learning. It involves identifying
that correctly classify instances into predefined categories
like decision trees or neural networks to search through po
hypotheses and select the best one.

Example	Sky	Air Temp	Humidity	Wind	Water	Forecast	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

- The given table represents a data set where Tom's enjoyment of his favorite water sports is recorded based on various weather conditions. The goal is to determine under what conditions Tom enjoys water sport

Attributes and Their Values

- **Sky**: Sunny, Rainy
- **Air Temp**: Warm, Cold
- **Humidity**: Normal, High
- **Wind**: Strong
- **Water**: Warm, Cool
- **Forecast**: Same, Change
- **Enjoy Sport**: Indicates whether Tom enjoys water sports under these conditions. Possible values: Yes, No

Learning Process

1. **Initialization:**
   - Start with the most specific hypothesis, which initially does not match any instances.
   - Example: $h = (\text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same})$
2. **Generalization:**
   - Update the hypothesis to generalize it as more examples are considered.
   - If the example is positive (EnjoySport = Yes), generalize the hypothesis to match the example.
3. **Updating Hypothesis:**
   - For each positive example, update the hypothesis to the least general generalization that still matches the example.
   - For each negative example, ensure the hypothesis does not match the negative example.

ep-by-Step Hypothesis Update

. **Initial Hypothesis**:

- $h = (\text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same})$

. **First Example (Sunny, Warm, Normal, Strong, Warm, Same, Yes)**:

- Hypothesis remains the same because it matches the example.
- $h = (\text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same})$

. **Second Example (Sunny, Warm, High, Strong, Warm, Same, Yes)**:

- Generalize the hypothesis to match both examples.
- $h = (\text{Sunny}, \text{Warm}, ?, \text{Strong}, \text{Warm}, \text{Same})$
- '?' means any value is acceptable for Humidity.

. **Third Example (Rainy, Cold, High, Strong, Warm, Change, No)**:

- Negative example, so no change to the hypothesis.

. **Fourth Example (Sunny, Warm, High, Strong, Cool, Change, Yes)**:

- Generalize the hypothesis further to include this example.
- $h = (\text{Sunny}, \text{Warm}, ?, \text{Strong}, ?, ?)$

Final Hypothesis

The final hypothesis represents the conditions under which Tom enjoys water sports:

- $(\text{Sunny}, \text{Warm}, ?, \text{Strong}, ?, ?)$

This means Tom enjoys water sports when:

- The sky is Sunny,
- The air temperature is Warm,
- The wind is Strong,
- Humidity, water temperature, and forecast can be any value.

- **Advantages**:
  - Generalization: Learns broad rules from specific examples.
  - Interpretability: Produces human-readable rules or models.

- **Disadvantages**:
  - Overfitting: Risk of overly complex models.
  - Requires Labeled Data: Needs a lot of labeled examples.

- **Applications**:
  - Email Spam Detection: Classifies emails as spam or not.
  - Medical Diagnosis: Predicts diseases from patient data.

Bayes Optimal Classifier

The Bayes’ Optimal Classifier is a theoretical model in ma
that makes predictions based on the highest posterior pro
it uses Bayes' theorem to combine prior knowledge with o
make the most accurate possible predictions.

Example: Spam Email Classification

Suppose we want to classify an email as "spam" or "not spam" based on certain features. Let's consider the features "contains the word 'offer'" and "contains the word 'win'".

Given Data:

- **Prior Probabilities:**
  - $P(\text{spam}) = 0.4$
  - $P(\text{not spam}) = 0.6$
- **Likelihoods:**
  - $P(\text{contains 'offer'}|\text{spam}) = 0.8$
  - $P(\text{contains 'win'}|\text{spam}) = 0.7$
  - $P(\text{contains 'offer'}|\text{not spam}) = 0.2$
  - $P(\text{contains 'win'}|\text{not spam}) = 0.1$
- **Feature Observation:**
  - The email contains both the words "offer" and "win".

Calculate Posterior Probabilities

1. **Calculate the joint likelihoods:**
   - For spam: $P(x|\text{spam}) = P(\text{contains 'offer'}|\text{spam}) \cdot P(\text{contains 'win'}|\text{spam})$

     $$P(x|\text{spam}) = 0.8 \cdot 0.7 = 0.56$$

   - For not spam: $P(x|\text{not spam}) = P(\text{contains 'offer'}|\text{not spam}) \cdot P(\text{contains 'win'}|\text{not spam})$

     $$P(x|\text{not spam}) = 0.2 \cdot 0.1 = 0.02$$

2. **Calculate the evidence:**
   - $P(x) = P(x|\text{spam}) \cdot P(\text{spam}) + P(x|\text{not spam}) \cdot P(\text{not spam})$

     $$P(x) = (0.56 \cdot 0.4) + (0.02 \cdot 0.6) = 0.224 + 0.012 = 0.236$$

3. **Calculate the posterior probabilities using Bayes' theorem:**
   - For spam:

     $$P(\text{spam}|x) = \frac{P(x|\text{spam}) \cdot P(\text{spam})}{P(x)} = \frac{0.56 \cdot 0.4}{0.236} \approx 0.949$$

   - For not spam:

     $$P(\text{not spam}|x) = \frac{P(x|\text{not spam}) \cdot P(\text{not spam})}{P(x)} = \frac{0.02 \cdot 0.6}{0.236} \approx 0.051$$

Since $P(\text{spam}|x) \approx 0.949$ and $P(\text{not spam}|x) \approx 0.051$, the Bayes Optimal Classifier will classify this email as "spam" because the posterior probability for "spam" is higher.

Posterior Probability Calculation:

- Uses Bayes' theorem to calculate the posterior probability of each class given the data.
- Formula: $P(C_i|x) = \frac{P(x|C_i) \cdot P(C_i)}{P(x)}$
  - $P(C_i|x)$: Posterior probability of class C_i given instance x
  - $P(x|C_i)$: Likelihood of instance x given class C_i
  - $P(C_i)$: Prior probability of class C_i
  - $P(x)$: Probability of instance x

Classification:

- Assigns the class label with the highest posterior probability to the instance.
- $C^* = \arg\max_{C_i} P(C_i|x)$

- **Advantages**:
  - Generalization: Learns broad rules from specific examples.
  - Interpretability: Produces human-readable rules or models.

- **Disadvantages**:
  - Overfitting: Risk of overly complex models.
  - Requires Labeled Data: Needs a lot of labeled examples.

- **Applications**:
  - Email Spam Detection: Classifies emails as spam or not.
  - Medical Diagnosis: Predicts diseases from patient data.

Naive Bayes Classifier

The Naive Bayes’ classifier is a simple and effective probabilistic cla
Bayes' Theorem. It assumes that the features are conditionally inde
the class label, which is often not true in practice but simplifies the c
significantly

Bayes' Theorem for Class "Yes" and "No" with Single and Multiple Features

1. Bayes' Theorem for Single Feature X and Class "Yes" (Y):

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

2. Bayes' Theorem for Multiple Features $X_1, X_2, \ldots, X_n$ and Class "Yes" (Y):

$$P(Y|X_1, X_2, \ldots, X_n) = \frac{P(X_1|Y) \cdot P(X_2|Y) \cdot P(X_3|Y) \cdots P(X_n|Y) \cdot P(Y)}{P(X_1) \cdot P(X_2) \cdot P(X_3) \cdots P(X_n)}$$

3. Bayes' Theorem for Single Feature X and Class "No" (N):

$$P(N|X) = \frac{P(X|N) \cdot P(N)}{P(X)}$$

4. Bayes' Theorem for Multiple Features $X_1, X_2, \ldots, X_n$ and Class "No" (N):

$$P(N|X_1, X_2, \ldots, X_n) = \frac{P(X_1|N) \cdot P(X_2|N) \cdot P(X_3|N) \cdots P(X_n|N) \cdot P(N)}{P(X_1) \cdot P(X_2) \cdot P(X_3) \cdots P(X_n)}$$

Overweight Due to Eating Habits or Medical Issues

Person	Eating Habits (Good/Bad)	Medical Issues (Yes/No)	Overweight (Yes/No)
1	Bad	No	Yes
2	Bad	No	No
3	Bad	Yes	Yes
4	Good	No	No
5	Good	Yes	Yes
6	Good	No	No
7	Bad	No	Yes
8	Good	Yes	Yes
9	Bad	No	No
10	Good	Yes	No

Person	Eating Habits (Good/Bad)	Medical Issues (Yes/No)	Overweight (Yes/No)
1	Bad	No	Yes
2	Bad	No	No
3	Bad	Yes	Yes
4	Good	No	No
5	Good	Yes	Yes
6	Good	No	No
7	Bad	No	Yes
8	Good	Yes	Yes
9	Bad	No	No
10	Good	Yes	No

Step 1: Prior Probability

- $P(\text{overweight} = \text{yes}) = \frac{5}{10} = 0.5$
- $P(\text{overweight} = \text{no}) = \frac{5}{10} = 0.5$

Step 2: Conditional Probability

	Overweight = Yes	Overweight = No
Eating Habits (Bad)	$\frac{3}{5}$	$\frac{2}{5}$
Eating Habits (Good)	$\frac{2}{5}$	$\frac{3}{5}$
Medical Issues (Yes)	$\frac{2}{5}$	$\frac{1}{5}$
Medical Issues (No)	$\frac{3}{5}$	$\frac{4}{5}$

Bayes' Theorem for Class "Yes" and "No" with Single and Multiple Features

1. Bayes' Theorem for Single Feature X and Class "Yes" (Y):

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

2. Bayes' Theorem for Multiple Features $X_1, X_2, \ldots, X_n$ and Class "Yes" (Y):

$$P(Y|X_1, X_2, \ldots, X_n) = \frac{P(X_1|Y) \cdot P(X_2|Y) \cdot P(X_3|Y) \cdots P(X_n|Y) \cdot P(Y)}{P(X_1) \cdot P(X_2) \cdot P(X_3) \cdots P(X_n)}$$

3. Bayes' Theorem for Single Feature X and Class "No" (N):

$$P(N|X) = \frac{P(X|N) \cdot P(N)}{P(X)}$$

4. Bayes' Theorem for Multiple Features $X_1, X_2, \ldots, X_n$ and Class "No" (N):

$$P(N|X_1, X_2, \ldots, X_n) = \frac{P(X_1|N) \cdot P(X_2|N) \cdot P(X_3|N) \cdots P(X_n|N) \cdot P(N)}{P(X_1) \cdot P(X_2) \cdot P(X_3) \cdots P(X_n)}$$

Step 3: Posterior Probability Calculation

Given a person with bad eating habits and medical issues, we want to calculate:

1. **Probability of being overweight given bad eating habits and medical issues:**

$$P(\text{overweight} = \text{yes} | \text{bad eating habits, medical issues}) = P(\text{bad eating habits} | \text{yes}) \cdot P(\text{medical issues} | \text{yes}) \cdot P(\text{yes})$$

$$P(\text{overweight} = \text{yes} | \text{bad eating habits, medical issues}) = \frac{3}{5} \cdot \frac{2}{5} \cdot 0.5 = 0.3 \cdot 0.4 = 0.12$$

2. **Probability of not being overweight given bad eating habits and medical issues:**

$$P(\text{overweight} = \text{no} | \text{bad eating habits, medical issues}) = P(\text{bad eating habits} | \text{no}) \cdot P(\text{medical issues} | \text{no}) \cdot P(\text{no})$$

$$P(\text{overweight} = \text{no} | \text{bad eating habits, medical issues}) = \frac{2}{5} \cdot \frac{1}{5} \cdot 0.5 = 0.2 \cdot 0.2 = 0.02$$

Conclusion

Given the person with bad eating habits and medical issues, we compare the posterior probabilities:

- $P(\text{overweight} = \text{yes} | \text{bad eating habits, medical issues}) = 0.12$
- $P(\text{overweight} = \text{no} | \text{bad eating habits, medical issues}) = 0.02$

Since $0.12 > 0.02$, the person is more likely to be overweight given bad eating habits and medical issues.

Bayes' Theorem

Bayes' Theorem for a given class C_i and feature vector x is:

$$P(C_i|x) = \frac{P(x|C_i) \cdot P(C_i)}{P(x)}$$

Naive Bayes Assumption

The naive assumption of feature independence means that the joint probability $P(x|C_i)$ can be decomposed into the product of individual probabilities:

$$P(x|C_i) = \prod_{j=1}^{n} P(x_j|C_i)$$

Thus, the classifier can be written as:

$$P(C_i|x) \propto P(C_i) \cdot \prod_{j=1}^{n} P(x_j|C_i)$$

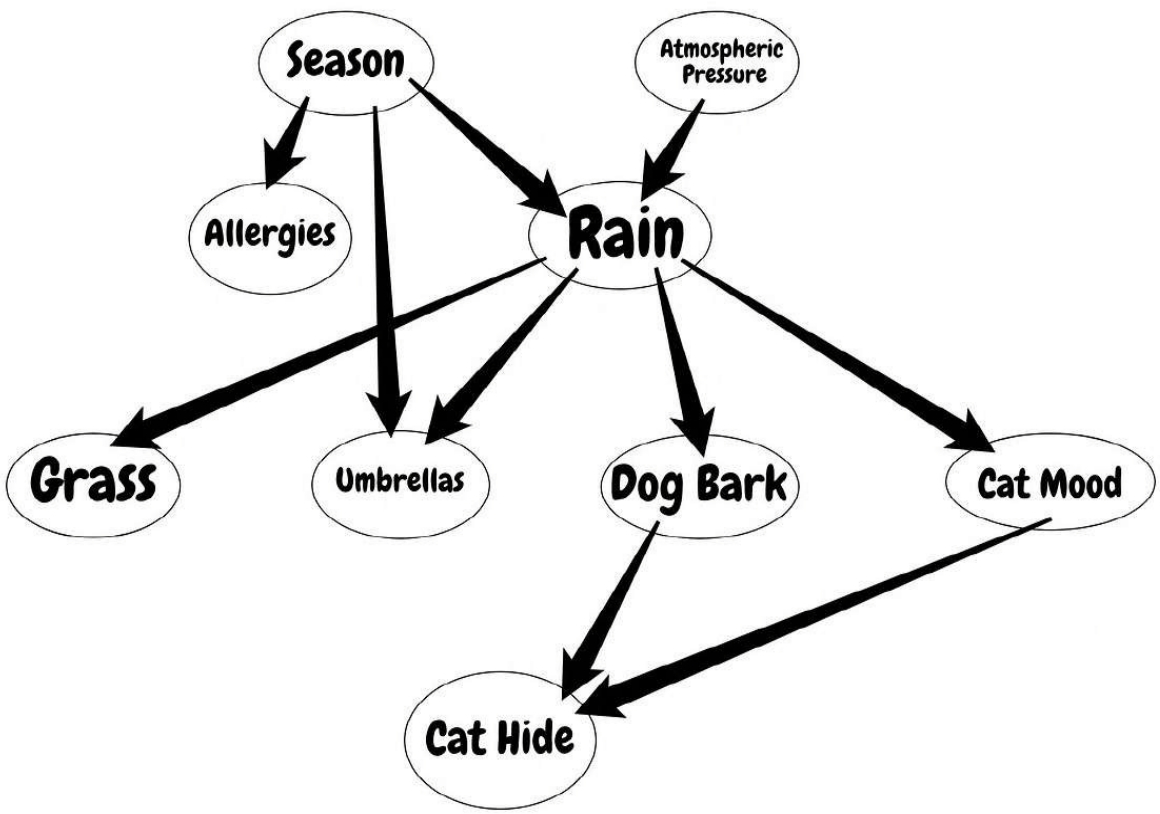
Classification Rule

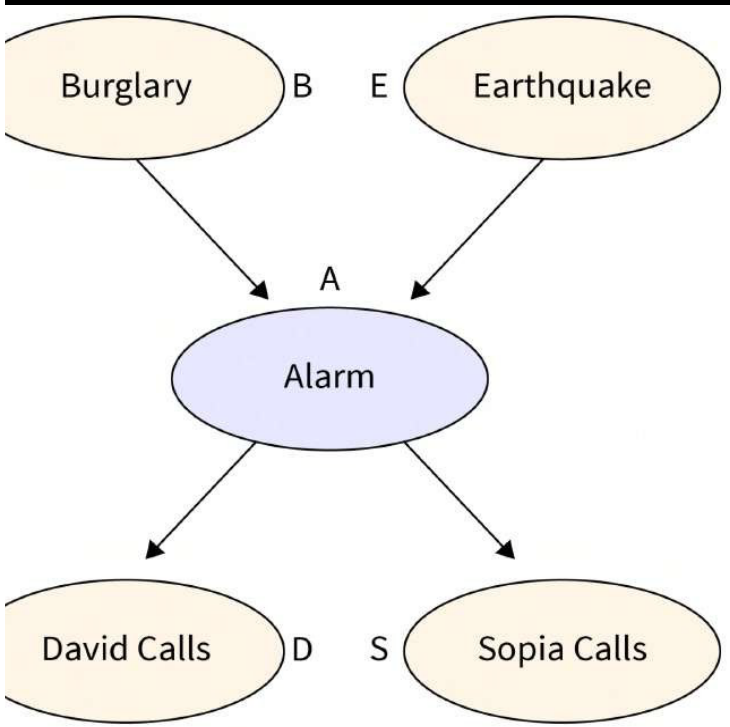
The Naive Bayes classifier assigns a class label $\hat{C}$ to a given instance x by choosing the class with the highest posterior probability:

$$\hat{C} = \arg\max_{C_i} P(C_i) \cdot \prod_{j=1}^{n} P(x_j|C_i)$$

Bayesian Belief Network (BBN)

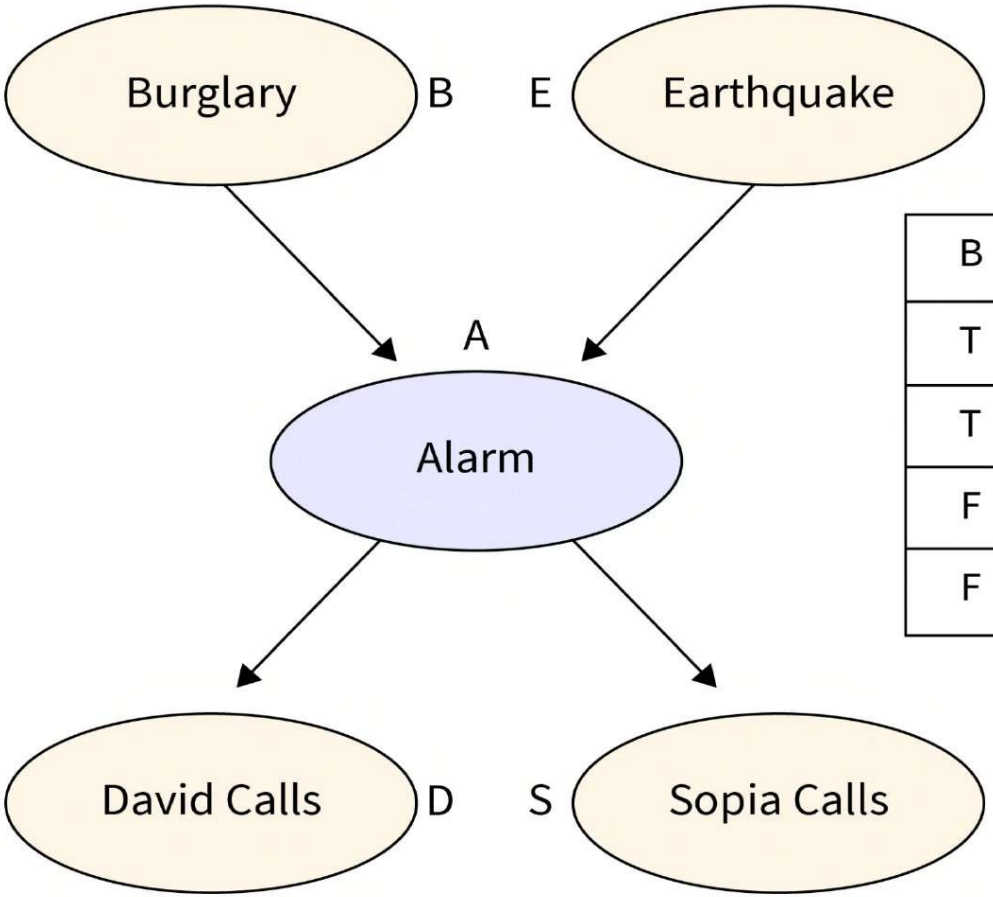
•A Bayesian Belief Network (BBN), also known as a Bayesian Network or Belief Network, is a graphical model that represents a set of variables and their conditional dependencies using a directed a cyclic graph (DAG).Each node in the network represents a variable, and each edge represents a conditional dependency between variables.





T	0.02
F	0.998

T	0.001
F	0.999



B	E	P(A=T)	P(A=F)
T	T	0.94	0.06
T	F	0.95	0.04
F	T	0.69	0.69
F	F	0.999	0.999

A	P(D=T)	P(D=F)
T	0.91	0.09
F	0.05	0.95

A	P(S=T)	P(S=F)
T	0.75	0.25
F	0.02	0.98

Key Components:

1. **Nodes**:
   - Represent random variables.
   - Example: Variables like "Smoking", "Cancer", "Coughing".
2. **Edges**:
   - Directed edges represent conditional dependencies.
   - Example: An edge from "Smoking" to "Cancer" indicates that smoking affects the probability of cancer.
3. **Conditional Probability Tables (CPTs)**:
   - Each node has a CPT that quantifies the effect of the parent nodes.
   - Example: The CPT for "Cancer" might show the probability of having cancer given whether or not a person smokes.

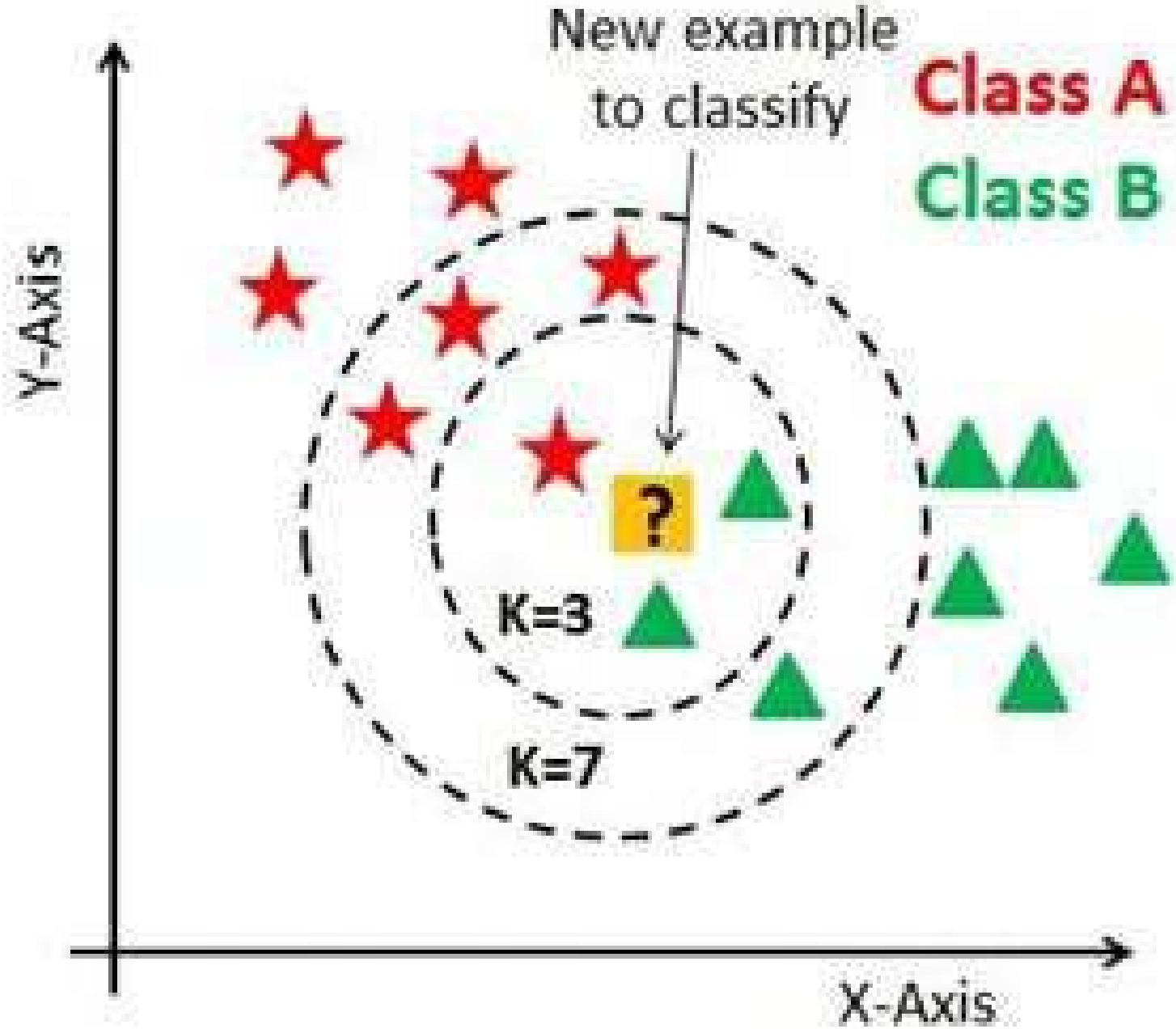
- **Advantages:**
  - **Modular Representation**: Each variable is conditionally independent of its non-descendants given its parents.
  - **EfficientComputation**: Supports efficient inference algorithms for calculating probabilities.
  - **Visualization**: Provides a clear visual representation of dependencies among variables.
- **Disadvantages:**
  - **Complexity**: Can become computationally intensive for large networks.
  - **Data Requirements**: Requires sufficient data to accurately estimate the CPTs.
  - **Design Effort**: Constructing a good network requires expert knowledge and effort.

- **Applications:**
  - **Medical Diagnosis**: Modeling diseases and symptoms to assist in diagnosis.
  - **Risk Assessment**: Evaluating risks in engineering and finance.
  - **Natural Language Processing**: Understanding dependencies between words and sentences.
  - **Decision Support Systems**: Providing recommendations based on Probabilistic reasoning.

Instance-Based Learning in Machine Learning

•**Historical Context**: Instance-based learning has been found
Recognition and classification tasks since the early days of AI.
algorithm, one of the simplest and most effective instance-bas
developed in the 1950s.

•**Definition**: Instance-based learning, also known as memory-
lazy learning, is a type of machine learning where the algorithm
of training data and uses them to make predictions on new dat
explicitly creating a model during the training phase, instance-
methods delay the generalization process until a query is mad
prediction phase.


New example
to classify
Class A
Class B
Y-Axis
?
K=3
K=7
X-Axis

Key Characteristics:

Lazy Learning: No explicit model is built during training. The algorithm simply stores the training instances. Generalization occurs at the time of prediction.

Similarity-Based: Predictions are made based on the similarity between the new data instance and stored training instances. Common similarity measures include Euclidean distance, Manhattan distance, and cosine similarity.

Local Decision Making: Decisions are made based on the local neighbor hood of the input instance. The algorithm considers only a subset of the training data (e.g., the k-nearest neighbours) to make a prediction.

Example Algorithms:

k-Nearest Neighbors (k-NN):

Algorithm: For a given input instance, the algorithm finds the k training Instances closest to the input and assigns the most common label (for classification) or averages the labels (for regression).

Application: Used in recommendation systems, image recognition, and anomaly detection.

Advantages of Instance-Based Learning:

- **Simplicity**:
  - Easy to implement and understand.
  - No need for a complex model training phase.
- **Adaptability**:
  - Can adapt quickly to new data without retraining the entire model.
  - Useful for applications where the data distribution changes overtime.
- **Performance**:
  - Effectiveforproblemswherethedecisionboundaryisirregularandcannot be easily captured by a global model.

Disadvantages of Instance-Based Learning:

Computational Cost:

High storage requirements since all training instances are retained.

Prediction can be slow for large data sets due to the need to compute distances to all stored instances.

Sensitivity to Noise:

Can be sensitive to noisy data, a so utliers can affect the prediction.

Feature Scaling:

Performance depends on the proper scaling of features, as different scales can distort distance calculations